

Do NOT write ANYTHING above this line!

1. (12 pts) Compute the following integrals

a) (4 pts)  $I_1 = \int \cos^3(x) \sin^2(x) dx = \int \cos^2 \sin^2 \underbrace{cdx}_{\textcircled{1} \frac{du}{u} = \cos x dx}$

$$= \int (1 - \sin^2) \sin^2 cdx = \int (1 - u^2) u^2 du = \int (u^2 - u^4) du$$

$$= \frac{1}{3} u^3 - \frac{1}{5} u^5 + c$$

$$I_1 = \frac{1}{3} \cos^3 x - \frac{1}{5} \cos^5 x + c$$

Not necessary

b) (4 pts)  $I_2 = \int_0^B x e^{Ax} dx = \left[ uv \right]_0^B - \int_0^B v du = \left[ \frac{x}{A} e^{Ax} \right]_0^B - \int_0^B \frac{1}{A} e^{Ax} dx$

Part:  $u = x \rightarrow du = dx$   
 $v = \frac{1}{A} e^{Ax} \rightarrow dv = e^{Ax} dx$

$$= \left[ \frac{x}{A} e^{Ax} - \frac{1}{A^2} e^{Ax} \right]_0^B = \left( \frac{B}{A} e^{AB} - \frac{1}{A^2} e^{AB} \right) - \left( 0 e^0 - \frac{1}{A^2} e^0 \right)$$

$$= \frac{B}{A} e^{AB} - \frac{1}{A^2} e^{AB} + \frac{1}{A^2}$$

$$I_2 = \left( B - \frac{1}{A} \right) \frac{e^{AB}}{A} + \frac{1}{A^2}$$

c) (4 pts)  $I_3 = \int \frac{7x+5}{x^2+x-2} dx$

Part:  $x^2+x-2 = (x+2)(x-1)$

$$\Rightarrow I_3 = \int \frac{A}{x+2} + \frac{B}{x-1} = \int \frac{A(x-1) + B(x+2)}{D}$$

$x=1$   $7(1)+5 = B(1+2) \Rightarrow 12 = 3B \Rightarrow \boxed{B=4}$   $\textcircled{1/2}$

$x=-2$   $7(-2)+5 = A(-2-1) \Rightarrow -9 = -3A \Rightarrow \boxed{A=3}$   $\textcircled{1/2}$

$$\Rightarrow I_3 = \int \frac{3}{x+2} + \int \frac{4}{x-1}$$

$$I_3 = 3 \ln|x+2| + 4 \ln|x-1| + c$$

Not necessary

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2. (6 pts) (a) Compute the Taylor polynomial of order 2 (i.e. 2nd degree polynomial) for  $f(x) = \sqrt{x-1}$  about  $x = 2$ .  
 (b) Sketch: (i)  $y = f(x)$  [using THICK SOLID line], (ii) the Taylor polynomial of order one (linear) [using THIN SOLID line], and (iii) the Taylor polynomial of order two [using DASHED line].

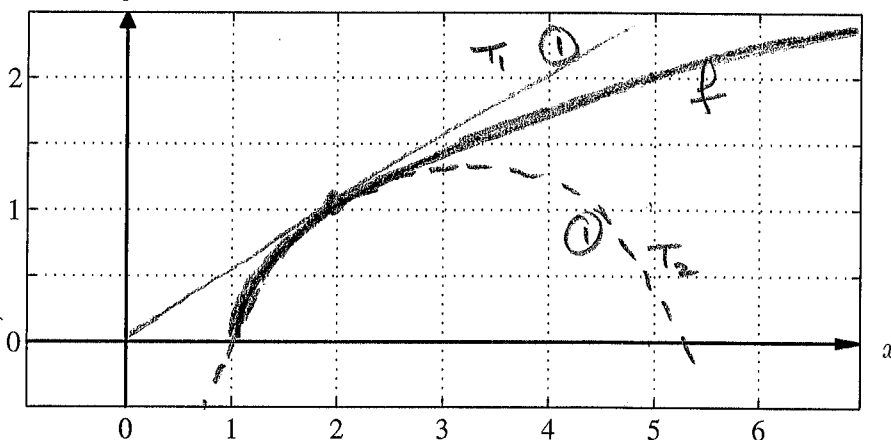
$$f(x) = \sqrt{x-1}, \quad f(2) = \sqrt{2-1} = 1$$

$$f'(x) = ((x-1)^{1/2})' = \frac{1}{2\sqrt{x-1}}, \quad f'(2) = \frac{1}{2\sqrt{2-1}} = \frac{1}{2}$$

$$f''(x) = \left(\frac{1}{2}(x-1)^{-1/2}\right)' = -\frac{1}{4}(x-1)^{-3/2} = -\frac{1}{4(x-1)^{3/2}}, \quad f''(2) = -\frac{1}{4 \cdot 1^{3/2}} = -\frac{1}{4}$$

① Taylor:  $\sum_{n=0}^{\infty} \frac{f^{(n)}(a)(x-a)^n}{n!}$

①  $T: 1 \frac{(x-2)^0}{0!} + \frac{1}{2} \frac{(x-2)^1}{1!} + \frac{-1/4}{2!} \frac{(x-2)^2}{2!} + \dots$



②  $f(x) \approx 1 + \frac{1}{2}(x-2) + \frac{(x-2)^2}{8}$

3. (6 pts) Application of series: A patient without any previous treatment is injected every 24 hrs with a drug. Immediately before each injection the concentration of the remaining drug in the patient is reduced by 80% from the previous day. Each injection increases the patient drug concentration by 2 mg/L. Denote by  $C_i$  the concentration immediately AFTER the  $i$ -th injection. Answer the following questions:

- (a) Write explicit expressions for  $C_1$ ,  $C_2$  and  $C_3$  (the concentrations after the first, second and third injections):

$$C_1 = 2$$

$$\textcircled{1} C_2 = \frac{20}{100} C_1 + 2 = \frac{2}{10} C_1 + 2 = \frac{1}{5} C_1 + 2 = \frac{1}{5} \cdot 2 + 2$$

$$\textcircled{1} C_3 = \frac{1}{5} C_2 + 2 = \frac{1}{5} \left[ \frac{1}{5} \cdot 2 + 2 \right] + 2 = \frac{1}{5} \cdot 2 + \frac{1}{5} \cdot 2 + 2$$

- (b) Write an explicit formula, using  $\Sigma$  notation, for the concentration  $C_i$  (just after the  $i$ -th injection) as a function of  $i$ .

$$\textcircled{2} \left| C_i = \sum_{n=1}^i \frac{1}{5^{n-1}} \cdot 2 \right|$$

- (c) If the patient remains indefinitely with this treatment, what would be the limiting value of the concentration?

$$C_{\infty} = \sum_{n=1}^{\infty} 2 \cdot \left(\frac{1}{5}\right)^{n-1} \quad \text{rk1}$$

$$\text{FIRST} \quad \frac{2}{1-r} = \frac{2}{1-1/5} = \frac{2}{4/5} = 5/2$$

$$\textcircled{2} \left| C_{\infty} = 5/2 \text{ mg/L} \right|$$

Do NOT wrt

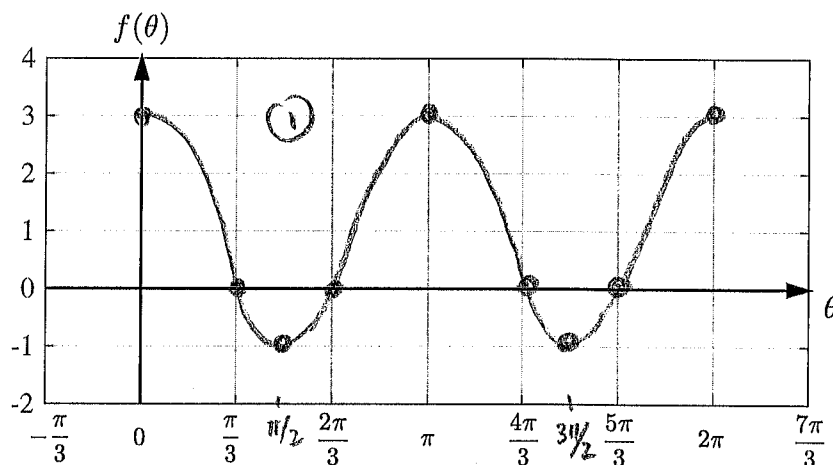
4. (5 pts) Let  $r = f(\theta) = 1 + 2 \cos(2\theta)$

a) Sketch the graph of  $r = f(\theta)$  for  $0 \leq \theta \leq 2\pi$ .  
in the Cartesian coordinates.

Indicate the values of  $\theta$  where:

- (i)  $f(\theta) = 0$ , and
- (ii) the points where  $f$  attains local maxima
- (iii) and local minima.

[Hint:  $\cos(2\pi/3) = -1/2 = \cos(4\pi/3)$ .]



(i) Zeros:

$\theta = \pi/3, 2\pi/3, 4\pi/3, 5\pi/3$

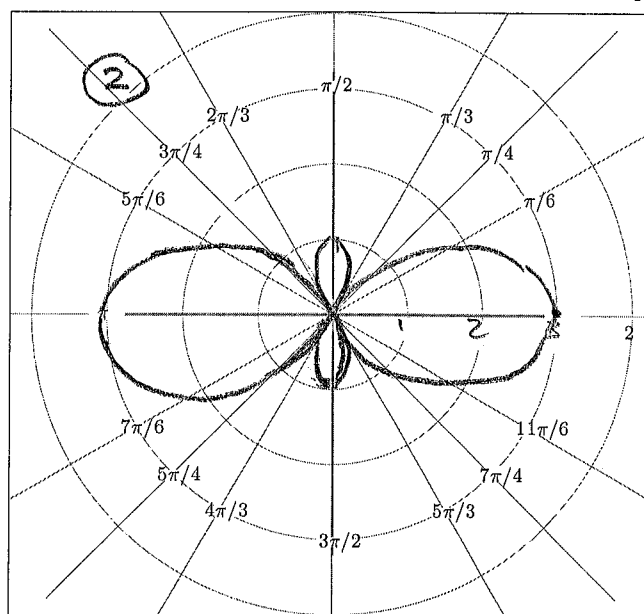
(ii) Maxima:

$\theta = 0, \pi, 2\pi$

(iii) Minima:

$\theta = \pi/2, 3\pi/2$

b) Sketch the graph of  $r = f(\theta)$  for  $r \in [0, 2\pi]$   
in the polar plot to the right.

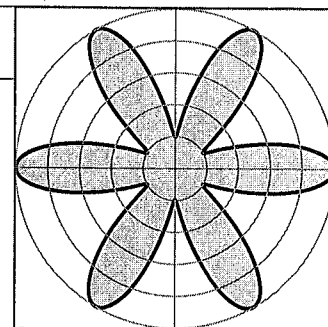


5. (5 pts) The graph to the right represents the silhouette of an asterisk that is described by the polar curve:  $r = 6 + 4 \cos(6\theta)$ .

a) Using the curve's **symmetry**, find an explicit integral for the AREA contained inside the curve. Do NOT compute the integral! [Note the factor of 2 in front of the integral!]

$$A = \int_0^{2\pi} \frac{1}{2} r^2(\theta) d\theta = \int_0^{2\pi} = 2 \int_0^{\pi} \frac{1}{2} r^2 d\theta$$

Area =  $2 \int_0^{\pi} \frac{1}{2} (6 + 4 \cos(6\theta))^2 d\theta$



b) Likewise, do the same for the total LENGTH of the curve. [Hint: note the factor of 2 in front of the integral!]

$$L = \int_0^{2\pi} \sqrt{r^2(\theta)^2 + (-24 \sin(6\theta))^2} d\theta = 2 \int_0^{\pi} \sqrt{(6 + 4 \cos(6\theta))^2 + (-24 \sin(6\theta))^2} d\theta$$

Length =  $2 \int_0^{\pi} \sqrt{(6 + 4 \cos(6\theta))^2 + (24 \sin(6\theta))^2} d\theta$

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6. (5 pts) Determine whether or not the following improper integral CONVERGES or DIVERGES.  
Please EXPLAIN and state any theorem/result that you are using. No explanation  $\Rightarrow$  NO POINTS !!!  
Compute the integral in case of convergence.

$$I_4 = \int_3^{\infty} \frac{7}{(x-2)^2} dx = \lim_{t \rightarrow \infty} \textcircled{1} \int_3^t \frac{7}{(x-2)^2} dx = \lim_{t \rightarrow \infty} \int_1^{t-2} \frac{7}{u^2} du$$

$$u = x-2 \quad \left| \begin{array}{l} x=3 \rightarrow u=1 \\ x=t \rightarrow u=t-2 \end{array} \right. \\ du = dx$$

$$= \lim_{t \rightarrow \infty} \int_1^{t-2} 7u^{-2} du = \lim_{t \rightarrow \infty} \left[ 7 \frac{u^{-1}}{-1} \right]_1^{t-2} = \lim_{t \rightarrow \infty} \left[ -\frac{7}{t-2} - \left(-\frac{7}{1}\right) \right] \textcircled{1}$$

$$= -\frac{7}{\infty-2} + 7 \therefore \text{CONV.}$$

OR Back to  $x$ :  $I_4 = \lim_{t \rightarrow \infty} \left[ 7 \frac{(x-2)^{-1}}{-1} \right]_3^t = -\frac{7}{\infty-2} - \frac{-7}{3-2} = 7$

Circle one:  $I_4$  Converges / Diverges

$I_4 = 7$  ①

7. (5 pts) Solve the following differential equation satisfying the given initial condition.  
(i) Give first the GENERAL solution and then (ii) the PARTICULAR solution satisfying for the initial condition.

$\frac{y'}{x^N} = B y$  with  $y(0) = A$  and where  $N$ ,  $A$ , and  $B$  are positive constants.

$y' = B x^N y$  SEP  $\frac{1}{y} dy = B x^N \Rightarrow \int \frac{1}{y} dy = \int B x^N dx$  ①

$\int \frac{1}{y} dy = B \int x^N dx \Rightarrow \ln|y| = B \frac{x^{N+1}}{N+1} + C$  ①

exp  $\Rightarrow |y| = e^{\frac{B x^{N+1}}{N+1} + C} = e^C e^{\frac{B x^{N+1}}{N+1}}$

$\Rightarrow y = \pm e^C e^{\frac{B x^{N+1}}{N+1}}$

OR  $y = K e^{\frac{B x^{N+1}}{N+1}}$

$y(0) = A$

$K e^0 = A \Rightarrow \boxed{K = A}$  ①

(i) General solution:

$y(x) = K e^{\frac{B x^{N+1}}{N+1}}$  ①

(ii) Particular solution:

$y(x) = A e^{\frac{B x^{N+1}}{N+1}}$  ①

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8. (5 pts) Determine the radius of convergence AND the interval of convergence of the following infinite series by using the RATIO TEST. Do NOT study the convergence at the end points. Show all your work!

$$\sum_{n=0}^{\infty} (-1)^n \frac{n^2}{4^n} (x+5)^n$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1 \Rightarrow \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} \frac{(n+1)^2}{4^{n+1}} (x+5)^{n+1}}{(-1)^n \frac{n^2}{4^n} (x+5)^n} \right| = \lim_{n \rightarrow \infty} \frac{(n+1)^2}{n^2} \frac{1}{4} |x+5|$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{(1 + 1/n)^2}{1} \frac{|x+5|}{4} = \frac{|x+5|}{4} < 1$$

$$\therefore \text{Ratio test} \Rightarrow |x+5| < 4 \quad R$$

$$\Rightarrow -4 < x+5 < 4$$

$$\Rightarrow \boxed{-9 < x < -1}$$

Radius of convergence:

4 (1)

Interval of convergence:

$-9 < x < -1$  (1)

9. (3 pts) Write an integral for the SURFACE AREA obtained by rotating about the line  $y = 1$  (note this is NOT one of the main axes!) the graph of  $y = f(x) = 2 + \sqrt{x-1}$  for  $1 \leq x \leq 5$ . DRAW A SKETCH. Do NOT compute this integral!

$$S = \int_{\text{arc}} ds = \int 2\pi r \cdot \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx$$

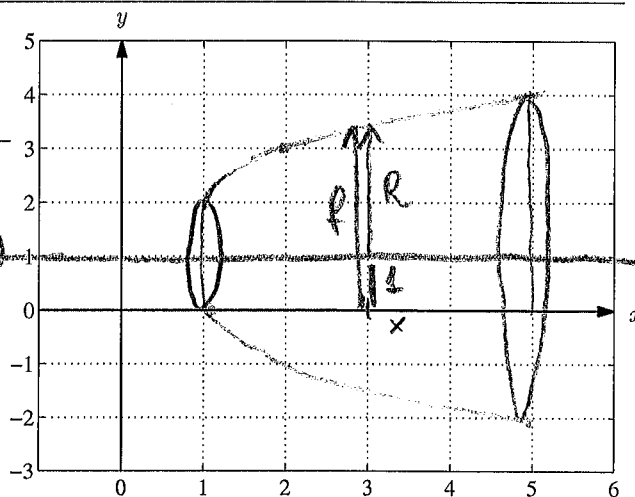
$$r = f(x) - 1 = 2 + \sqrt{x-1} - 1 = 1 + \sqrt{x-1} \quad (1)$$

$$f' = \frac{1}{2\sqrt{x-1}} \Rightarrow (f')^2 = \frac{1}{4(x-1)}$$

$$S = \int_1^5 2\pi (1 + \sqrt{x-1}) \sqrt{1 + \frac{1}{4(x-1)}} dx$$

(1) [No 1/2 pts!]

6



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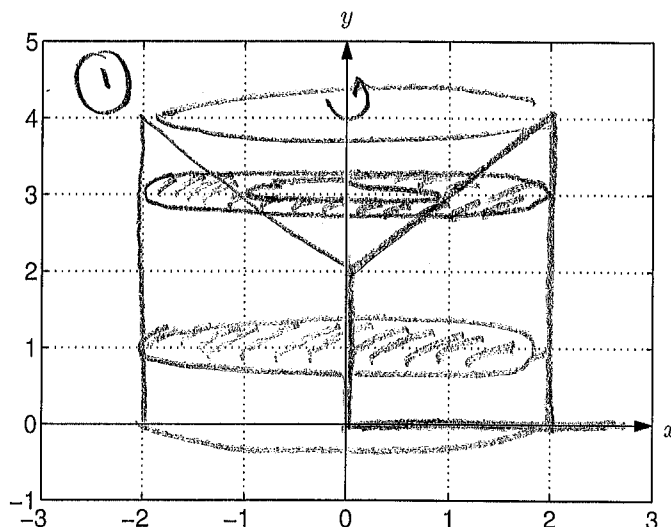
10. (6 pts) Using **WASHERS** write an integral (or integrals) for the volume generated by rotating about the  $y$ -axis the region in the first quadrant delimited by the following functions:  $x = 0$ ,  $x = 2$ ,  $y = 0$ , and  $y = x + 2$ . Sketch the functions, the region, the solid and a typical washer.

$$V = \int_{y_1}^{y_2} A(y) dy = \left| \int_0^2 \pi (r_o^2 - r_i^2) + \int_2^5 \pi (r_o^2 - r_i^2) \right| \quad (1)$$

$$0 < y < 2 \quad | \quad r_o = 2; \quad r_i = 0 \quad (1)$$

$$2 < y < 5 \quad | \quad r_o = 2; \quad r_i = x \quad \text{but } y = x + 2 \\ \Rightarrow \quad r_i = x = y - 2 \quad (1)$$

$$V_1 = \int_0^2 \pi (2^2 - 0^2) \\ + \int_2^5 \pi (2^2 - (y-2)^2)$$



[Do NOT compute any integrals!]

(2)

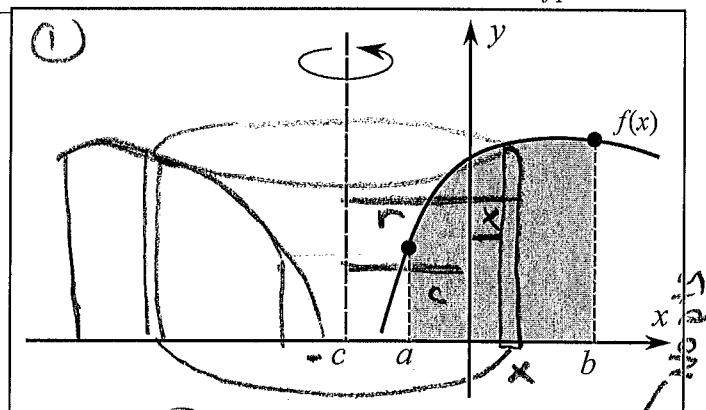
$$V_1 = \int_0^2 \pi 4 dy + \int_2^5 \pi (4 - (y-2)^2) dy$$

11. (5 pts) Using the method of volumes by **SHELLS** write an integral for the solid generated by rotating about the  $x = -c$  axis the shaded region shown in the plot. Note that the axis of rotation is off-axis! Sketch the solid and a typical shell.

$$V_{\text{SHELLS}} = \int_{x_1}^{x_2} 2\pi r \cdot h \, dx$$

$$r = x + c \quad | \quad h = f(x)$$

$$\therefore V = \int_a^b 2\pi (x+c) f(x) dx$$



$$V_2 = \int_a^b 2\pi (x+c) f(x) dx$$

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12. (4 pts) Write an integral for the length of the curve depicted in the graph and defined by the graph of  $y = f(x)$  for  $1 \leq x \leq 3$  using (i) an integral over  $x$  AND (ii) an integral over  $y$ .

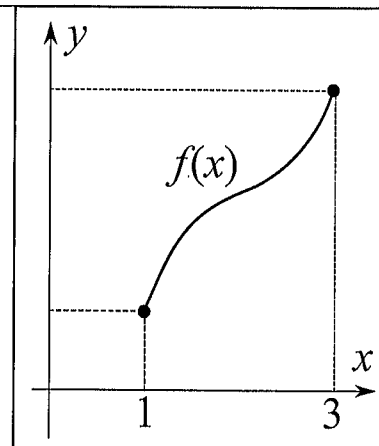
$L = \int ds$   $ds = \sqrt{dx^2 + dy^2} < \sqrt{1 + (f'(x))^2}$

(i) Using integral over  $x$ :

$L_x = \int_{\boxed{1}}^{\boxed{3}} \boxed{\sqrt{1 + (f'(x))^2}} dx$

(ii) Using integral over  $y$ :

$L_y = \int_{\boxed{f(1)}}^{\boxed{f(3)}} \boxed{\sqrt{1 + ((f^{-1}(y)))'^2}} dy$



Use this space for scratch work

① [No Y2 pts]

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1. (12 pts) Compute the following integrals

a) (4 pts)  $I_1 = \int \cos^2(x) \sin^3(x) dx = \int c^2 s^2 \underline{3dx} = \int c^2 (1-c^2) 3dx$   
 $du = \sin x dx$   
 $u = -\cos x$   
 $= \int (-u)^2 (1-u^2) du$   
 $= \int u^2 (1-u^2) du = \int (u^2 - u^4) du = \frac{1}{3} u^3 - \frac{1}{5} u^5 + C$

$I_1 = \frac{1}{3} \sin^3 x - \frac{1}{5} \sin^5 x + C$  Not necessary

b) (4 pts)  $I_2 = \int_0^D x e^{Bx} dx = [uv]_0^D - \int_0^D v du = \left[ \frac{x}{B} e^{Bx} \right]_0^D - \int_0^D \frac{1}{B} e^{Bx} dx$

part:  $u = x \rightarrow du = dx$   
 $v = \frac{1}{B} e^{Bx} \rightarrow v = e^{Bx} dx$

$= \left[ \frac{x}{B} e^{Bx} - \frac{1}{B^2} e^{Bx} \right]_0^D = \left( \frac{D}{B} e^{BD} - \frac{1}{B^2} e^{BD} \right) - \left( \frac{0}{B} e^0 - \frac{1}{B^2} e^0 \right)$   
 $= \frac{D}{B} e^{BD} - \frac{1}{B^2} e^{BD} + \frac{1}{B^2}$  OR

$I_2 = (D - \frac{1}{B}) \frac{e^{BD}}{B} + \frac{1}{B^2}$

c) (4 pts)  $I_3 = \int \frac{8x-1}{x^2-x-2} dx$

partial fractions:  $x^2-x-2 = (x-2)(x+1)$

$I_3 = \int \frac{A}{x-2} + \frac{B}{x+1} = \int \frac{A(x+1) + B(x-2)}{D}$

$\underline{x = -1} \quad 8x-1 = 8(-1)-1 = -9 \quad \Rightarrow \quad B(-1-2) = -3B \Rightarrow \underline{B = 3}$   
 $\underline{x = 2} \quad 8x-1 = 15 = A(2+1) = 3A \quad \Rightarrow \quad \underline{A = 5}$

$\therefore I_3 = \int \frac{5}{x-2} + \frac{3}{x+1}$

$I_3 = 5 \ln|x-2| + 3 \ln|x+1| + C$  Not necessary



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2. (6 pts) (a) Compute the Taylor polynomial of order 2 (i.e. 2nd degree polynomial) for  $f(x) = \sqrt{x-2}$  about  $x = 3$ .  
 (b) Sketch: (i)  $y = f(x)$  [using THICK SOLID line], (ii) the Taylor polynomial of order one (linear) [using THIN SOLID line], and (iii) the Taylor polynomial of order two [using DASHED line].

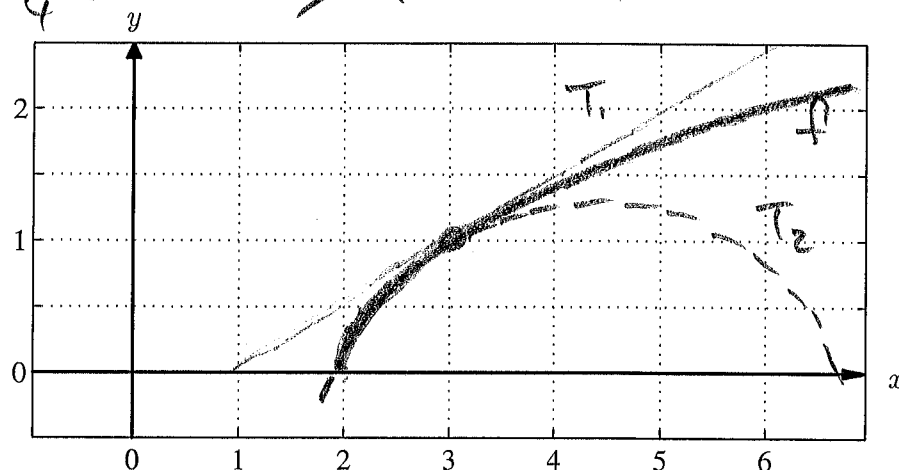
$$f(x) = \sqrt{x-2}, \quad f(3) = \sqrt{3-2} = \sqrt{1} = 1$$

$$f'(x) = \frac{1}{2\sqrt{x-2}}, \quad f'(3) = \frac{1}{2\sqrt{1}} = \frac{1}{2}$$

$$f''(x) = \left(\frac{1}{2}(x-2)^{-1/2}\right)' = -\frac{1}{4}(x-2)^{-3/2}, \quad f''(3) = -\frac{1}{4}(1)^{-3/2} = -\frac{1}{4}$$

$$\text{Taylor: } \sum_{n=0}^{\infty} \frac{f^{(n)}(a)(x-a)^n}{n!}$$

$$T: 1 \frac{(x-3)^0}{0!} + \frac{1}{2} \frac{(x-3)^1}{1!} + \frac{1}{4} \frac{(x-3)^2}{2!}$$



$$f(x) \approx 1 + \frac{1}{2}(x-3) - \frac{1}{4}(x-3)^2$$

3. (6 pts) Application of series: A patient without any previous treatment is injected every 24 hrs with a drug. Immediately before each injection the concentration of the remaining drug in the patient is reduced by 60% from the previous day. Each injection increases the patient drug concentration by 3 mg/L. Denote by  $C_i$  the concentration immediately AFTER the  $i$ -th injection. Answer the following questions:

- (a) Write explicit expressions for  $C_1$ ,  $C_2$  and  $C_3$  (the concentrations after the first, second and third injections):

$$C_1 = 3$$

$$C_2 = \frac{40}{100} C_1 + 3 = \frac{4}{10} C_1 + 3 = \frac{2}{5} C_1 + 3 = \frac{2}{5} \cdot 3 + 3$$

$$C_3 = \frac{2}{5} C_2 + 3 = \frac{2}{5} \left[ \frac{2}{5} \cdot 3 + 3 \right] + 3 = \left( \frac{2}{5} \right)^2 \cdot 3 + \left( \frac{2}{5} \right) \cdot 3 + \left( \frac{2}{5} \right)^0 \cdot 3$$

- (b) Write an explicit formula, using  $\Sigma$  notation, for the concentration  $C_i$  (just after the  $i$ -th injection) as a function of  $i$ .

$$C_i = \sum_{n=0}^i \left( \frac{2}{5} \right)^{n-1} \cdot 3$$

- (c) If the patient remains indefinitely with this treatment, what would be the limiting value of the concentration?

$$C_{\infty} = \sum_{n=0}^{\infty} \left( \frac{2}{5} \right)^{n-1} \cdot 3 \stackrel{\text{FIRST}}{=} \frac{3}{1 - 2/5} = \frac{3}{3/5} = 5$$

$$\therefore \boxed{C_{\infty} = 5 \text{ mg/L}}$$

Do NOT write

4. (5 pts) Let  $r = f(\theta) = 1 + 2 \cos(2\theta)$

a) Sketch the graph of  $r = f(\theta)$  for  $0 \leq \theta \leq 2\pi$  in the Cartesian coordinates.

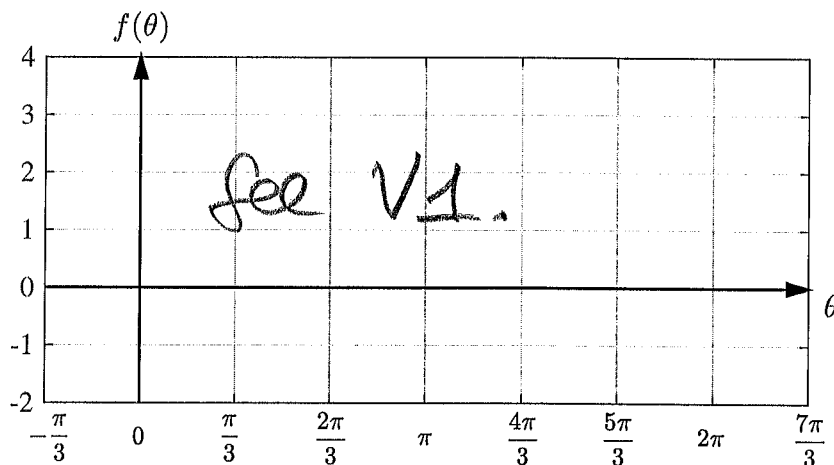
Indicate the values of  $\theta$  where:

(i)  $f(\theta) = 0$ , and

(ii) the points where  $f$  attains local maxima

(iii) and local minima.

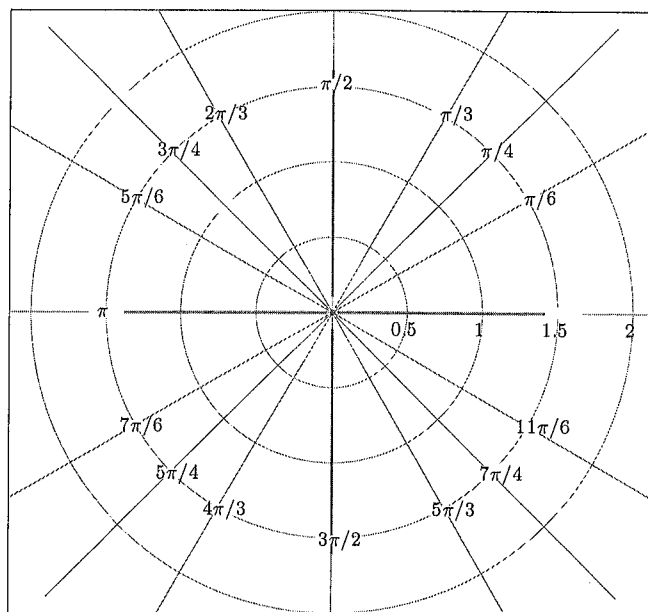
[Hint:  $\cos(2\pi/3) = -1/2 = \cos(4\pi/3)$ .]



(i) Zeros:  $\theta =$

(ii) Maxima:  $\theta =$

(iii) Minima:  $\theta =$

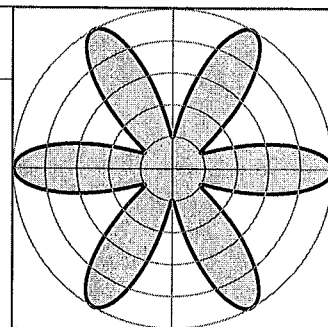


b) Sketch the graph of  $r = f(\theta)$  for  $r \in [0, 2\pi]$  in the polar plot to the right.

5. (5 pts) The graph to the right represents the silhouette of an asterisk that is described by the polar curve:  $r = 3 + 2 \cos(6\theta)$ .

a) Using the curve's **symmetry**, find an explicit integral for the AREA contained inside the curve. Do NOT compute the integral! [Note the factor of 2 in front of the integral!]

$$A = \int_0^{2\pi} \frac{1}{2} r^2 d\theta = \int_0^{2\pi} \frac{1}{2} (3 + 2 \cos(6\theta))^2 d\theta$$



$$\text{Area} = 2 \int_0^{\pi} \frac{1}{2} (3 + 2 \cos(6\theta))^2 d\theta$$

b) Likewise, do the same for the total LENGTH of the curve. [Hint: note the factor of 2 in front of the integral!]

$$L = \int_0^{2\pi} \sqrt{r^2 + (r')^2} d\theta = 2 \int_0^{\pi} \sqrt{(3 + 2 \cos(6\theta))^2 + (-12 \sin(6\theta))^2} d\theta$$

$$\text{Length} = 2 \int_0^{\pi} \sqrt{(3 + 2 \cos(6\theta))^2 + (12 \sin(6\theta))^2} d\theta$$

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6. (5 pts) Determine whether or not the following improper integral CONVERGES or DIVERGES.

Please EXPLAIN and state any theorem/result that you are using. No explanation  $\Rightarrow$  NO POINTS !!!

Compute the integral in case of convergence.

$$I_4 = \int_2^{\infty} \frac{5}{(x-1)^2} dx = \lim_{t \rightarrow \infty} \int_2^t \frac{5}{(x-1)^2} dx = \lim_{t \rightarrow \infty} \int_1^{t-1} \frac{5}{u^2} du$$

$$u = x-1 \mid x=2 \rightarrow u=1$$

$$du = dx \mid x=t \rightarrow u=t-1$$

$$= \lim_{t \rightarrow \infty} \int_1^{t-1} 5u^{-2} du = \lim_{t \rightarrow \infty} \left[ \frac{5u^{-1}}{-1} \right]_1^{t-1} = \lim_{t \rightarrow \infty} \left[ -\frac{5}{t-1} + \frac{5}{2-1} \right]$$

$$= -\frac{5}{\infty-1} + 5 \therefore \text{conv.}$$

OR Back to  $x$ :  $I_4 = \lim_{t \rightarrow \infty} \left[ \frac{5(x-1)^{-1}}{-1} \right]_2^t = \frac{-5}{t-1} - \frac{-5}{2-1} = 5$

Circle one:  $I_4$ : Converges / Diverges

$I_4 = 5$

7. (5 pts) Solve the following differential equation satisfying the given initial condition.

(i) Give first the GENERAL solution and then (ii) the PARTICULAR solution satisfying for the initial condition.

$\frac{y'}{x^N} = Ay$  with  $y(0) = B$  and where  $N$ ,  $A$ , and  $B$  are positive constants.

See V1 w/ B  $\rightarrow$  A

(i) General solution:  $y(x) =$

(ii) Particular solution:  $y(x) =$

Do NOT write ANYTHING above this line!

8. (5 pts) Determine the radius of convergence AND the interval of convergence of the following infinite series by using the **RATIO TEST**. Do NOT study the convergence at the end points. Show all your work!

$$\sum_{n=0}^{\infty} (-1)^n \frac{n^2}{7^n} (x+3)^n$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1 \Rightarrow \lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} (n+1)^2 (x+3)^{n+1}}{(-1)^n n^2 (x+3)^n} \right| = \lim_{n \rightarrow \infty} \frac{(n+1)^2}{n^2} \frac{|x+3|}{7}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{n^2 (1 + 1/n)^2}{n^2} \frac{|x+3|}{7} = (1 + 0)^2 \frac{|x+3|}{7} = \frac{|x+3|}{7}$$

$$\therefore \text{Ratio test} \Rightarrow |x+3| < 7 \leftarrow R$$

$$\Rightarrow -7 < x+3 < 7$$

$$\begin{aligned} -3 \\ \Rightarrow -10 < x < 4 \end{aligned}$$

Radius of convergence:

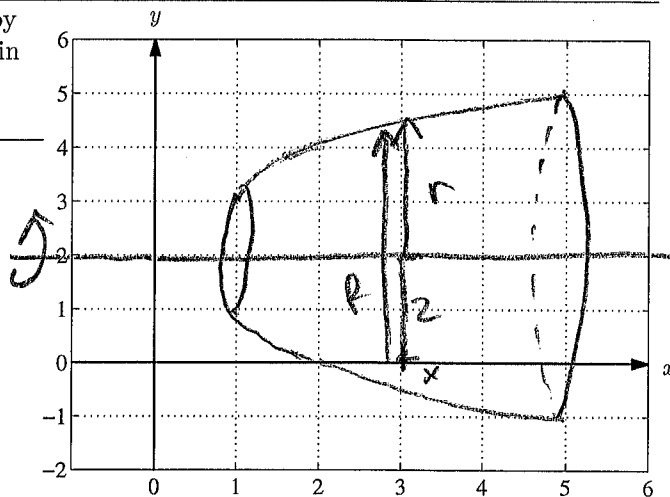
$$R=7$$

Interval of convergence:

$$-10 < x < 4$$

9. (3 pts) Write an integral for the SURFACE AREA obtained by rotating about the line  $y = 2$  (note this is NOT one of the main axes!) the graph of  $y = f(x) = 3 + \sqrt{x-1}$  for  $1 \leq x \leq 5$ . DRAW A SKETCH. Do NOT compute this integral!

$$\begin{aligned} S &= \int_{arc} ds = \int 2\pi r \sqrt{1 + (f')^2} dx \\ r &= f(x) - 2 = 3 + \sqrt{x-1} - 2 = 1 + \sqrt{x-1} \\ f' &= \frac{1}{2\sqrt{x-1}} \Rightarrow (f')^2 = \frac{1}{4(x-1)} \end{aligned}$$



$$S = \int_1^5 2\pi (1 + \sqrt{x-1}) \sqrt{1 + \frac{1}{4(x-1)}} dx$$

Do NOT write ANYTHING above this line!

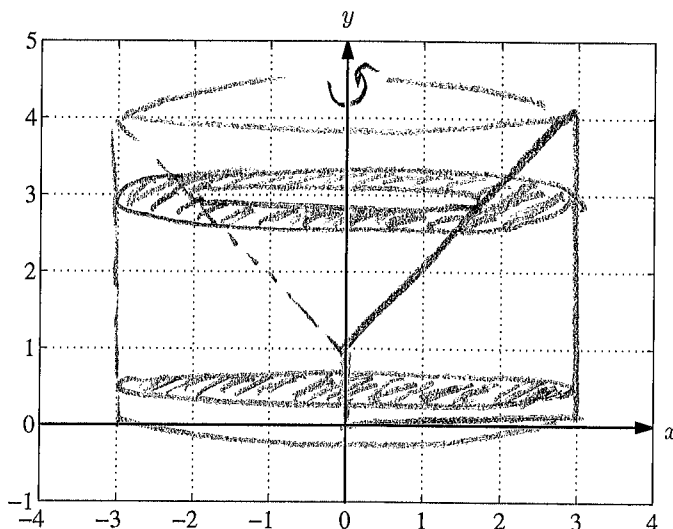
10. (6 pts) Using **WASHERS** write an integral (or integrals) for the volume generated by rotating about the  $y$ -axis the region in the first quadrant delimited by the following functions:  $x = 0$ ,  $x = 3$ ,  $y = 0$ , and  $y = x + 1$ . Sketch the functions, the region, the solid and a typical washer.

$$V = \int_{y_1}^{y_2} A dy = \int_0^1 \pi (R_o^2 - r_i^2) + \int_1^4 \pi (R_o^2 - r_i^2)$$

$$0 < y < 1 \quad | R_o = 3, r_i = 0 |$$

$$1 < y < 4 \quad | R_o = 3, r_i = x \text{ \& } y = x + 1 \\ \Rightarrow | r_i = x = y - 1 |$$

$$\therefore V_1 = \int_0^1 \pi (3^2 - 0^2) dy + \int_1^4 \pi (3^2 - (y-1)^2) dy$$



[Do NOT compute any integral!]

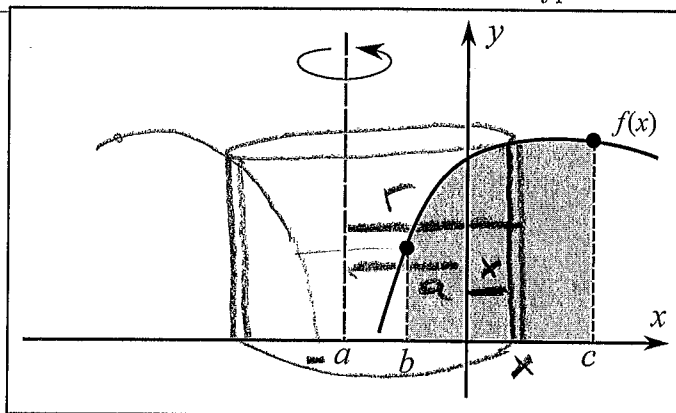
$$V_1 = \int_0^1 \pi 9 dy + \int_1^4 \pi (9 - (y-1)^2) dy$$

11. (5 pts) Using the method of volumes by **SHELLS** write an integral for the solid generated by rotating about the  $x = -a$  axis the shaded region shown in the plot. Note that the axis of rotation is off-axis! Sketch the solid and a typical shell.

$$V_{\text{SHELLS}} = \int \text{arc. height} \\ = \int_{x_1}^{x_2} 2\pi r \cdot h \cdot dx$$

$$\text{r} | r = a + x \quad | \quad h = f(x)$$

$$\therefore V = \int_b^c 2\pi (a+x) f(x) dx$$



$$V_2 = \int_b^c 2\pi (a+x) f(x) dx \quad d \quad \times$$

Do NOT write ANYTHING above this line!

12. (4 pts) Write an integral for the length of the curve depicted in the graph and defined by the graph of  $y = f(x)$  for  $2 \leq x \leq 4$  using (i) an integral over  $x$  AND (ii) an integral over  $y$ .

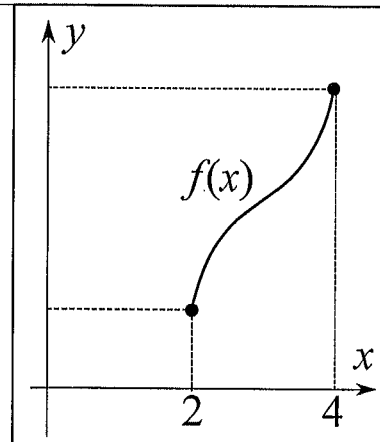
$$L = \int ds \quad ; \quad ds = \sqrt{dx^2 + dy^2} \quad \Leftrightarrow \quad \sqrt{1 + (f'(x))^2} \quad \text{or} \quad \sqrt{1 + (f'(y))^2}$$

(i) Using integral over  $x$ :

$$L_x = \int_{\boxed{2}}^{\boxed{4}} \boxed{\sqrt{1 + (f'(x))^2}} \, dx$$

(ii) Using integral over  $y$ :

$$L_y = \int_{\boxed{f(2)}}^{\boxed{f(4)}} \boxed{\sqrt{1 + (f'(y))^2}} \, dy$$



Use this space for scratch work